LLM trading agents:

We evaluate FINCON against four LLM agents in the context of stock trading.

- **GENERAL-PURPOSE GENERATIVE AGENTS – GA:**The generative AI agent by Park et al. [20], originally intended to simulate realistic human behavior and make everyday decisions, has been adapted here for specific stock trading tasks. This agent's architecture includes a memory module that employs recency, relevance, and importance metrics to extract pivotal memory events for informed decision-making. However, it does not provide a layered memory module to effectively differentiate the time sensitivities unique to various types of financial data. Additionally, although it features a profiling module to define agent attributes like professional background, the model does not specify the agent's persona. In our experiments, we modified the original prompt template created by Park et al., which was intended for general daily tasks, to suit financial investment tasks.
- **FINGPT:** A novel open-source LLM framework specialized for converting incoming textual and numeric information into informed financial decision-making, introduced by Yang et al[31]. It claims superiority over the traditional buy-and-hold strategy.
- **FINMEM:**FINMEM employs a specialized profiling module and self-adaptive risk settings for enhanced market robustness. Its memory module integrates working memory and layered long-term memory, enabling effective data processing. This allows FINMEM to leverage market insights and improve trading decisions [32].
- **FINAGENT:**FINAGENT developed upon FINMEM, which leverages the use of tool-using capabilities of LLMs to incorporate multi-modal financial data [33]. It claims an further improved trading performance on single asset trading (stocks and cryptocurrencies).

DRL trading agents:

As the FINMEM is practiced and examined on the basis of single stock trading and discrete trading actions, we choose three advanced DRL algorithms fitting into the same scenarios according to the previous and shown expressive performance in the work of Liu et al [98, 99]. The DRL training agents only take numeric features as inputs.

- **Advantage Actor-Critic (A2C):** A2C ([100]) is applied to optimize trading actions in the financial environment. It operates by simultaneously updating both the policy (actor) and the value (critic) functions, providing a balance between exploration and exploitation.
- **Proximal Policy Optimization (PPO):** PPO ([101]) is employed in stock trading due to its stability and efficiency. One salient advantage of PPO is that it maintains a balance between exploration and exploitation by bounding the policy update, preventing drastic policy changes.
- **Deep Q-Network (DQN):** DQN ([102]) is an adaptation of Q-learning, that can be used to optimize investment strategies. Unlike traditional Q-learning that relies on a tabular approach for storing Q-values, DQN generalizes Q-value estimation across states using deep learning, making it more scalable for complex trading environments.

A.12 Portfolio Management

Markowitz Portfolio Selection [1]: introduced by Harry Markowitz in 1952, is a framework for constructing portfolios that optimize expected return for a given level of risk or minimize risk for a given level of expected return. This method uses expected returns, variances, and covariances of asset returns to determine the optimal asset allocation, thereby balancing risk and return through diversification.

FinRL-A2C [48]: is an RL algorithm proposed to address single stock trading and portfolio optimization problems in Liu et al.. The RL models make trading decisions (i.e., portfolio weights) based on the observation of previous market conditions and the brokerage information of the RL agents. The implementation of this algorithm [2] is provided and is used as baselines in our study.

Equal-Weighted ETF [103]: is a portfolio giving equal allocation to all stocks, similar to a buy-and-hold strategy in single-stock trading, can provide a benchmark on market trends.

[2] `https://github.com/AI4Finance-Foundation/FinRL-Meta`

A.13 Ranking Metrics for Procedural Memory in FINCON

Upon receiving an investment inquiry, each agent in FINCON retrieves the top-K pivotal memory events from its procedural memory, where K is a hyperparameter. These events are selected based on their information retrieval score. For any given memory event E, its information retrieval score γ^E is defined by

$$\gamma^E = S^E_{\text{Relevancy}} + S^E_{\text{Importance}} \tag{10}$$

which is adpated from Park et al [20] but with modified relevancy and importance computations, and is scaled to $[0,1]$ before summing up. Upon the arrival of a trade inquiry P in processing memory event E via LLM prompts, the agent computes the relevancy score $S^E_{\text{Relevancy}}$ that measures the *cosine similarity* between the embedding vectors of the memory event textual content $\mathbf{m_E}$ and the LLM prompt query $\mathbf{m_P}$, which is defined as follows:

$$S^E_{\text{Relevancy}} = \frac{\mathbf{m_E} \cdot \mathbf{m_P}}{\|\mathbf{m_E}\|_2 \times \|\mathbf{m_P}\|_2} \tag{11}$$

Note that the LLM prompt query inputs trading inquiry and trader characteristics. On the other hand, the importance score $S^E_{\text{Importance}}$ is inversely correlates with the time gap between the inquiry and the event's memory timestamp $\delta t = t_{\text{P}} - t_E$, mirroring Ebbinghaus's forgetting curve [104]. More precisely, if we denote the initial score value of memory event v^E and degrading ratio $\theta \in (0,1)$, then the importance score is computed via

$$S^E_{\text{Importance}} = v^E \times \theta^{\delta t} \tag{12}$$

Note that the ratio θ measures the diminishing importance of an event over time, which is inspired by design of [20]. But in our design, the factors of recency and importance are handled by one equation. Different agents in FINCON admit different choices of $\{v^E, \theta\}$ for memory event E.

Additionally, an access counter function facilitates memory event augmentation, so that critical events impacting trading decisions can be augmented by FINCON, while trivial events are gradually faded. This is achieved by using the LLM validation tool Guardrails AI [105] to track critical memory ID. A memory ID deemed critical to investment gains receives $+5$ to its importance score $S^E_{\text{Importance}}$. This access counter implementation enables FINCON to capture and prioritize crucial events based on type and retrieval frequency.

A.14 Detailed Configurations in Experiments

The training period was chosen to account for the seasonal nature of corporate financial reporting and the duration of data retention in FINCON's memory module. The selected training duration ensures the inclusion of at least one publication cycle of either Form 10-Q, ECC, or Form 10-K. This strategy ensures that the learned conceptualized investment guidance considers a more comprehensive scope of factors. Additionally, the training duration allowed FINCON sufficient time to establish inferential links between financial news, market indicators, and stock market trends, thereby accumulating substantial experience. Furthermore, we set the number of top memory events retrieved for each agent at 5. We ran FINCON. The reported performance outcomes are based on the setting that achieved the highest cumulative return during the testing phase.

To maintain consistency in the comparison, the training and testing phases for the other three LLM-based agents were aligned with those of FINMEM. For parameters of other LLM-based agents that are not encompassed by FINMEM's configuration, they were kept in accordance with their original settings as specified in their respective source codes.

FINCON's performance was benchmarked against that of the most effective comparative model, using Cumulative Return and Sharpe Ratio as the primary evaluation metrics. The statistical significance of FINCON's superior performance was ascertained through the non-parametric Wilcoxon signed-rank test, which is particularly apt for the non-Gaussian distributed data.

A.15 FINCON performance on extreme market conditions

To further illustrate the robustness of FINCON's performance, we assess its effectiveness in two distinct scenarios: (1) a single-asset trading task using TSLA and (2) a portfolio management task involving a combination of TSLA, MSFT, and PFE. Our evaluation focuses on key financial metrics, including Cumulative Returns (CRs), Sharpe Ratios (SRs), and Maximum Drawdown (MDD). The training period spanned from January 17, 2022, to March 31, 2022, while the testing phase covered April 1, 2022, to October 15, 2022. This specific timeframe was chosen due to the elevated levels of the CBOE Volatility Index (VIX), which averaged above 20, signaling greater market volatility during these months.

As demonstrated in Table 7 and Figure 12, FINCON is the sole agent system to achieve positive Cumulative Returns (CRs) and Sharpe Ratios (SRs) in single stock trading tasks. Regarding portfolio management tasks, the results of all baselines (four benchmarks) are detailed in Table 8 and Figure 13. In these comparisons, FINCON consistently attained the highest scores in the primary performance metrics.

Models	CR % ↑	SR↑	MDD %↓
B&H	-56.738	-1.625	52.077
FinCon	22.460	0.695	45.215
GA	-51.251	-1.547	48.763
FinGPT	-20.035	-0.805	32.199
FinMem	-47.809	-1.549	49.560
FinAgent	-31.119	-1.933	33.224
A2C	-73.251	-2.142	56.998
PPO	-78.007	-2.284	59.003
DQN	-8.452	-1.328	8.463

Table 7: Key performance comparison for single asset trading under the high volatility condition using TSLA as an example. FinCon leads all performance metrics.

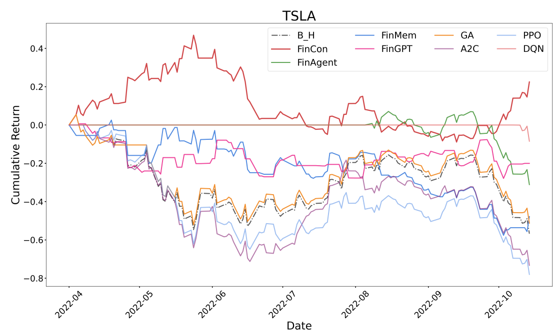

Figure 12: CR changes over time across all the strategies under the high volatility condition using TSLA as an example of the single asset trading task.

Models	CR % ↑	SR↑	MDD %↓
FinCon	-8.429	-0.294	26.176
Markowitz MV	-28.996	-1.805	31.831
FinRL-A2C	-15.932	-1.195	21.569
Equal-Weighted ETF	-28.008	-1.731	30.070

Table 8: Key performance comparison among all portfolio management strategies of Portfolio1 under the high volatility condition. FinCon leads all performance metrics.

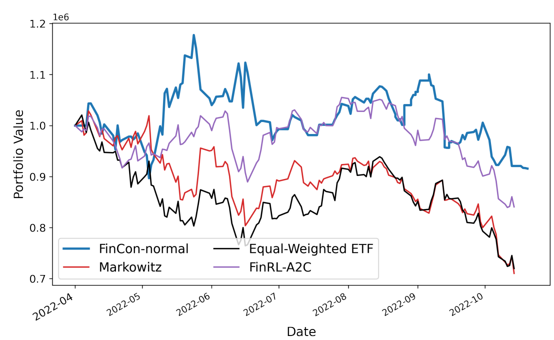

Figure 13: Portfolio1 value changes over time for all the strategies under the high volatility condition.